

Why Data Cleaning Matters: Methods and Tools in R

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2025-11-26

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(gt)
library(readxl)
theme_set(theme_minimal())
Mappe1 <- read_excel("Mappe1.xlsx")
```

Agenda

- What is it good for?
- Overview of methods & tools
- Exercise

What is Data Cleaning Good for?

Key benefits:

- **Accuracy** of results by removing errors, duplicates or other inconsistencies
- **Consistency** across datasets makes it possible to merge and compare datasets without conflict
- Handling **missing values** to prevent biased outcomes
- Better modeling and **predictions** when using statistical models

see Kabacoff (2024), Lee (2025)

Overview

Package	Function	Use
dplyr	select	select variables/columns
dplyr	filter	select observations/rows
dplyr	mutate	transform or recode variables
dplyr	summarize	summarize data
dplyr	group_by	identify subgroups for further processing

see Nordmann et al. (2022), Kabacoff (2024)

dataset starwars

```
starwars %>%  
  head(10)
```

```
starwars %>%  
  head(10)
```

```
# A tibble: 10 x 14
```

	name	height	mass	hair_color	skin_color	eye_color	birth_year	sex	gender
	<chr>	<int>	<dbl>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	Luke Sk~	172	77	blond	fair	blue	19	male	mascu~
2	C-3P0	167	75	<NA>	gold	yellow	112	none	mascu~
3	R2-D2	96	32	<NA>	white, bl~	red	33	none	mascu~

```

4 Darth V~      202    136 none      white      yellow      41.9 male  mascu~
5 Leia Or~      150     49 brown     light      brown       19  fema~ femin~
6 Owen La~      178    120 brown, gr~ light      blue       52  male  mascu~
7 Beru Wh~      165     75 brown     light      blue       47  fema~ femin~
8 R5-D4         97     32 <NA>      white, red red       NA   none  mascu~
9 Biggs D~      183     84 black     light      brown       24  male  mascu~
10 Obi-Wan~     182     77 auburn, w~ fair       blue-gray   57  male  mascu~
# i 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>

```

select()

- keeps only the columns you need, removes irrelevant ones

```

starwars %>%
select(name,species,contains("color")) %>%
head(10)

```

```

starwars %>%
select(name,species,contains("color")) %>%
head(10)

```

A tibble: 10 x 5

	name <chr>	species <chr>	hair_color <chr>	skin_color <chr>	eye_color <chr>
1	Luke Skywalker	Human	blond	fair	blue
2	C-3PO	Droid	<NA>	gold	yellow
3	R2-D2	Droid	<NA>	white, blue	red
4	Darth Vader	Human	none	white	yellow
5	Leia Organa	Human	brown	light	brown
6	Owen Lars	Human	brown, grey	light	blue
7	Beru Whitesun Lars	Human	brown	light	blue
8	R5-D4	Droid	<NA>	white, red	red
9	Biggs Darklighter	Human	black	light	brown
10	Obi-Wan Kenobi	Human	auburn, white	fair	blue-gray

filter()

- keeps only rows that meet certain conditions

```
starwars %>%
  select(name, height, species, eye_color) %>%
  filter(species == "Human" &
         height < 200 &
         eye_color %in% c("blue", "brown")) %>%
  head(9)
```

```
starwars %>%
  select(name, height, species, eye_color) %>%
  filter(species == "Human" &
         height < 200 &
         eye_color %in% c("blue", "brown")) %>%
  head(9)
```

```
# A tibble: 9 x 4
  name          height species eye_color
<chr>          <int> <chr>   <chr>
1 Luke Skywalker    172 Human   blue
2 Leia Organa       150 Human   brown
3 Owen Lars         178 Human   blue
4 Beru Whitesun Lars 165 Human   blue
5 Biggs Darklighter 183 Human   brown
6 Anakin Skywalker  188 Human   blue
7 Wilhuff Tarkin    180 Human   blue
8 Han Solo          180 Human   brown
9 Boba Fett         183 Human   brown
```

mutate()

- creates new variables or modifies existing ones

```
starwars %>%
  select(name, height, mass, species) %>%
  mutate(height = height/100) %>%
  head(10)
```

```
starwars %>%
  select(name, height, mass, species) %>%
  mutate(height = height/100) %>%
  head(10)
```

```
# A tibble: 10 x 4
```

	name	height	mass	species
	<chr>	<dbl>	<dbl>	<chr>
1	Luke Skywalker	1.72	77	Human
2	C-3P0	1.67	75	Droid
3	R2-D2	0.96	32	Droid
4	Darth Vader	2.02	136	Human
5	Leia Organa	1.5	49	Human
6	Owen Lars	1.78	120	Human
7	Beru Whitesun Lars	1.65	75	Human
8	R5-D4	0.97	32	Droid
9	Biggs Darklighter	1.83	84	Human
10	Obi-Wan Kenobi	1.82	77	Human

mutate()

- creates new variables or modifies existing ones

```
starwars %>%
  select(name, height, mass, species) %>%
  mutate(height_2 = height/100) %>%
  head(10)
```

```
starwars %>%
  select(name, height, mass, species) %>%
  mutate(height_2 = height/100) %>%
  head(10)
```

```
# A tibble: 10 x 5
```

	name	height	mass	species	height_2
	<chr>	<int>	<dbl>	<chr>	<dbl>
1	Luke Skywalker	172	77	Human	1.72
2	C-3P0	167	75	Droid	1.67
3	R2-D2	96	32	Droid	0.96
4	Darth Vader	202	136	Human	2.02
5	Leia Organa	150	49	Human	1.5
6	Owen Lars	178	120	Human	1.78
7	Beru Whitesun Lars	165	75	Human	1.65
8	R5-D4	97	32	Droid	0.97
9	Biggs Darklighter	183	84	Human	1.83
10	Obi-Wan Kenobi	182	77	Human	1.82

mutate() and recode ()

- replace or rename categories for consistency

```
starwars %>%
  select(name, contains("color"), species) %>%
  mutate(species = recode(species,
                          "Droid" = "Robot")) %>%
  head(10)
```

```
starwars %>%
  select(name, contains("color"), species) %>%
  mutate(species = recode(species,
                          "Droid" = "Robot")) %>%
  head(10)
```

A tibble: 10 x 5

	name	hair_color	skin_color	eye_color	species
	<chr>	<chr>	<chr>	<chr>	<chr>
1	Luke Skywalker	blond	fair	blue	Human
2	C-3PO	<NA>	gold	yellow	Robot
3	R2-D2	<NA>	white, blue	red	Robot
4	Darth Vader	none	white	yellow	Human
5	Leia Organa	brown	light	brown	Human
6	Owen Lars	brown, grey	light	blue	Human
7	Beru Whitesun Lars	brown	light	blue	Human
8	R5-D4	<NA>	white, red	red	Robot
9	Biggs Darklighter	black	light	brown	Human
10	Obi-Wan Kenobi	auburn, white	fair	blue-gray	Human

missing data

```
starwars %>%
  select(name, gender, hair_color, height) %>%
  head(10)
```

```
starwars %>%
  select(name, gender, hair_color, height) %>%
  head(10)
```

```
# A tibble: 10 x 4
```

	name <chr>	gender <chr>	hair_color <chr>	height <int>
1	Luke Skywalker	masculine	blond	172
2	C-3PO	masculine	<NA>	167
3	R2-D2	masculine	<NA>	96
4	Darth Vader	masculine	none	202
5	Leia Organa	feminine	brown	150
6	Owen Lars	masculine	brown, grey	178
7	Beru Whitesun Lars	feminine	brown	165
8	R5-D4	masculine	<NA>	97
9	Biggs Darklighter	masculine	black	183
10	Obi-Wan Kenobi	masculine	auburn, white	182

missing data - na.omit()

- drops all rows with missing values

```
starwars %>%  
  select(name, gender, hair_color, height) %>%  
  na.omit()  
head(10)
```

```
starwars %>%  
  select(name, gender, hair_color, height) %>%  
  na.omit()
```

```
# A tibble: 72 x 4
```

	name <chr>	gender <chr>	hair_color <chr>	height <int>
1	Luke Skywalker	masculine	blond	172
2	Darth Vader	masculine	none	202
3	Leia Organa	feminine	brown	150
4	Owen Lars	masculine	brown, grey	178
5	Beru Whitesun Lars	feminine	brown	165
6	Biggs Darklighter	masculine	black	183
7	Obi-Wan Kenobi	masculine	auburn, white	182
8	Anakin Skywalker	masculine	blond	188
9	Wilhuff Tarkin	masculine	auburn, grey	180
10	Chewbacca	masculine	brown	228

```
# i 62 more rows
```

```
head(10)
```

```
[1] 10
```

missing data - filter()

- shows all rows with missing value

```
starwars %>%  
  select(name, gender, hair_color, height) %>%  
  filter(!complete.cases(.))
```

```
starwars %>%  
  select(name, gender, hair_color, height) %>%  
  filter(!complete.cases(.))
```

```
# A tibble: 15 x 4
```

	name <chr>	gender <chr>	hair_color <chr>	height <int>
1	C-3PO	masculine	<NA>	167
2	R2-D2	masculine	<NA>	96
3	R5-D4	masculine	<NA>	97
4	Greedo	masculine	<NA>	173
5	Jabba Desilijic Tiure	masculine	<NA>	175
6	Jek Tono Porkins	<NA>	brown	180
7	Arvel Crynyd	masculine	brown	NA
8	Gregar Typho	<NA>	black	185
9	Cordé	<NA>	brown	157
10	Sly Moore	<NA>	none	178
11	Finn	masculine	black	NA
12	Rey	feminine	brown	NA
13	Poe Dameron	masculine	brown	NA
14	BB8	masculine	none	NA
15	Captain Phasma	feminine	none	NA

missing data - drop.na()

- remove rows only if specific columns have missing values

```
starwars %>%
  select(name, gender, hair_color, height) %>%
  filter(!complete.cases()) %>%
  drop_na(height)
head(10)
```

```
starwars %>%
  select(name, gender, hair_color, height) %>%
  filter(!complete.cases()) %>%
  drop_na(height)
```

A tibble: 9 x 4

	name	gender	hair_color	height
	<chr>	<chr>	<chr>	<int>
1	C-3PO	masculine	<NA>	167
2	R2-D2	masculine	<NA>	96
3	R5-D4	masculine	<NA>	97
4	Greedo	masculine	<NA>	173
5	Jabba Desilijic Tiure	masculine	<NA>	175
6	Jek Tono Porkins	<NA>	brown	180
7	Gregar Typho	<NA>	black	185
8	Cordé	<NA>	brown	157
9	Sly Moore	<NA>	none	178

```
head(10)
```

```
[1] 10
```

missing data - replace_na()

- fills missing values with defaults

```
starwars %>%
  select(name, gender, hair_color, height) %>%
  filter(!complete.cases()) %>%
  mutate(hair_color = replace_na(hair_color, "none"))
```

```
starwars %>%
  select(name, gender, hair_color, height) %>%
  filter(!complete.cases()) %>%
  mutate(hair_color = replace_na(hair_color, "none"))
```

```
# A tibble: 15 x 4
```

	name <chr>	gender <chr>	hair_color <chr>	height <int>
1	C-3P0	masculine	none	167
2	R2-D2	masculine	none	96
3	R5-D4	masculine	none	97
4	Greedo	masculine	none	173
5	Jabba Desilijic Tiure	masculine	none	175
6	Jek Tono Porkins	<NA>	brown	180
7	Arvel Crynyd	masculine	brown	NA
8	Gregar Typho	<NA>	black	185
9	Cordé	<NA>	brown	157
10	Sly Moore	<NA>	none	178
11	Finn	masculine	black	NA
12	Rey	feminine	brown	NA
13	Poe Dameron	masculine	brown	NA
14	BB8	masculine	none	NA
15	Captain Phasma	feminine	none	NA

group_by () and summarise ()

- organise data into groups (grouped operations) and collapse groups into summary statistics (e.g., mean, count, sum)

```
starwars %>%  
  group_by(species) %>%  
  summarise(avg_mass = mean(mass, na.rm = TRUE))
```

```
starwars %>%  
  group_by(species) %>%  
  summarise(avg_mass = mean(mass, na.rm = TRUE))
```

```
# A tibble: 38 x 2
```

	species <chr>	avg_mass <dbl>
1	Aleena	15
2	Besalisk	102
3	Cerean	82
4	Chagrian	NaN
5	Clawdite	55
6	Droid	69.8

```
7 Dug          40
8 Ewok         20
9 Geonosian    80
10 Gungan      74
# i 28 more rows
```

```
head(10)
```

```
[1] 10
```

Learning my Doing

Task: What is the average female and male height (in m) and mass in the “starwars” data set?

```
# install.packages("tidyverse")
library(tidyverse)
view(starwars)
```

Try to use the functions: - select() - filter() - mutate() - drop_na() - group_by() - summarise()

Possible Solution

```
starwars %>%
  select(sex, height, mass) %>%
  filter(sex == "male" | sex == "female") %>%
  mutate(height = height/100) %>%
  drop_na() %>%
  group_by(sex) %>%
  summarise('Average height' = (mean(height)),
            'Average mass' = mean(mass))
```

```
starwars %>%
  select(sex, height, mass) %>%
  filter(sex == "male" | sex == "female") %>%
  mutate(height = height/100) %>%
  drop_na() %>%
  group_by(sex) %>%
```

```
summarise('Average height' = (mean(height)),  
          'Average mass' = mean(mass))
```

```
# A tibble: 2 x 3  
  sex    `Average height` `Average mass`  
  <chr>          <dbl>         <dbl>  
1 female          1.72           54.7  
2 male            1.78           80.2
```

Thank you for listening!

References

- Kabacoff, R. (2024). *Modern data visualization with r*. Chapman; Hall/CRC.
- Lee, S. (2025). *The ultimate guide to data cleaning in r*. <https://www.numberanalytics.com/blog/ultimate-data-cleaning-r>
- Nordmann, E., McAleer, P., Toivo, W., Paterson, H., & DeBruine, L. M. (2022). Data visualization using r for researchers who do not use r. *Advances in Methods and Practices in Psychological Science*, 5(2), n.p.